Abhinav Singh

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EDUCATION

Bennett University, Greater Noida, UP *Expected Jun 2028*
B.Tech – Computer Science & Engineering

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Solidity, Rust, CSS
Blockchain / Web3: Solidity, ERC-721, Hardhat, ethers.js, IPFS, Circom, Anchor (Solana), SP1 zkVM
Frontend / Backend: React.js, Next.js, Node.js, Tailwind CSS, FastAPI
Databases: PostgreSQL, MySQL, MongoDB
Frameworks / Libraries: OpenZeppelin, wagmi, RainbowKit, Prisma, snarkjs, Foundry, NumPy

PROJECT EXPERIENCE

ZK-Whistleblower – Anonymous Blockchain Reporting Platform (GitHub | Live Demo) *Feb 2026 – Present*

- Built a full-stack anonymous whistleblowing platform using ZK-SNARKs (Groth16, Circom 2), allowing organisation insiders to prove membership — via Poseidon-hashed Merkle trees and nullifier hashes — without revealing their identity.
- Developed Solidity smart contracts (Hardhat 3 + OpenZeppelin) for on-chain ZK proof verification, Merkle root management, and tamper-proof IPFS-CID evidence storage.
- Architected a Next.js 15 monorepo (Admin + Reporter apps) with wagmi v2 + RainbowKit and client-side Groth16 proof generation in the browser via snarkjs.
- Wrote 13 tests covering valid proofs, nullifier replay attacks, revoked roots, and fake proofs — all using real ZK proof generation with zero mocks.

zk-ML Agent Vault – Verifiable DeFi Portfolio Agent (GitHub | Live Demo) *May 2026 – Present*

- Built a verifiable DeFi portfolio agent running a 2-layer MLP inside the SP1 zkVM (Rust + SP1 SDK v6.2.3); generates Groth16 proofs that trading decisions were computed honestly before on-chain execution.
- End-to-end ZK pipeline: NumPy-trained weights baked into the SP1 guest via include_bytes!, with inputs and predicted action committed to public IO inside the zkVM.
- Wrote MLAgentVault.sol (Solidity 0.8.20) for on-chain SP1 proof verification and vault management; tested with Foundry unit + fuzz tests against a mock SP1 verifier.

ERC-721 NFT Smart Contract – Scalable NFT System (GitHub) *Mar 2026 – Apr 2026*

- Implemented a production-grade ERC-721 contract with owner-minted and public fee-based minting, max supply cap, and auto-incrementing token IDs using OpenZeppelin.
- Added security features: ReentrancyGuard for payable flows, Pausable controls, and owner access control with a public mint toggle; integrated IPFS-based per-token metadata URI storage.
- Deployed and verified on Ethereum Sepolia testnet with interaction scripts for ownerOf() and tokenURI() reads.

Sol-Uptime – Decentralised Uptime Monitor on Solana (GitHub) *Aug 2025 – Nov 2025*

- Collaborated on a decentralised ping/uptime monitoring system built as a Solana on-chain program using the Anchor framework in Rust, deployed to Solana Devnet.
- Contributed to the Next.js frontend integrating @solana/web3.js wallet UI and Anchor-generated client code, demonstrating multi-chain competency across EVM and non-EVM ecosystems.

CERTIFICATIONS

- NPTEL – Blockchain and Its Applications
- NPTEL – Data Structures and Algorithms Design
- Coursera – The Bits and Bytes of Computer Networking
- Coursera – Operating Systems

ACHIEVEMENT

NPTEL Blockchain and Its Applications — Elite + Silver Medal, Top 5% nationally.